

EmbodyCraft: Exploring Haptic Embodied Experiences for Reflective Practice in Throwing Clay

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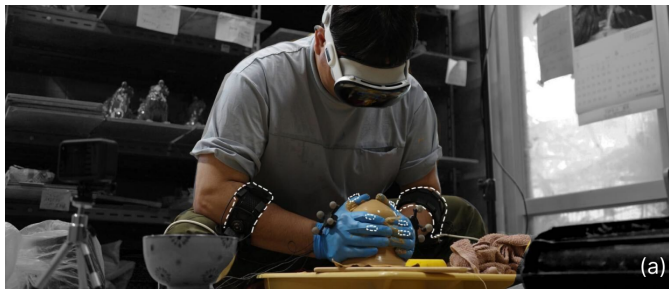


Figure 1: (a) Recording materials for reflection - Force-Myography(FMG) device, accelerometers, and cameras were used to record materials for reflection. (b) Reflecting on actions - Haptic devices were used for post-action reflection.

Abstract

Reflection is a crucial process for advancing craftsmanship. Much of craft knowledge is tacit, which cannot be easily conveyed through language, therefore leaving post-action reflection largely relying on personal recollection. Existing methods such as visualizing physical performance via video-recording have been used to support the reflection of craft practitioners, but they struggle to capture the subtle bodily actions and perceptions central to craft. To address this gap, we collaborated with seasoned potters to develop a system called *EmbodyCraft*. It can record muscle activity and fingertip vibration when the potter touches the clay, which then translates into tactile stimuli for post-action reflection. Comparing findings from two workshops with potters of varying experience levels, we observed that supplementing conventional video-based reflection with our haptic cues helped reveal subtle force patterns and correct misconceptions about both one's own and others' actions. These

preliminary results suggest a potential for deeper skill understanding during reflection, as well as more nuanced communication in skill-based discussions.

CCS Concepts

• **Human-centered computing** → **Haptic devices; Empirical studies in HCI.**

Keywords

Embodiment; Haptic Interface; Skin-Stretch Stimulation; Reflection; Craft Skill

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1 Introduction

Reflection is a critical activity in craft practices, which enables practitioners to assess “*what has been done and what remains undone*” [34], thereby improving their skills and deepening their understanding of the craft itself. However, reflection in craft practices faces unique challenges as most of the knowledge is tacit [13, 18, 26, 35].

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As Polanyi described, “*we can know more than we can tell*”, tacit knowledge is deeply embedded in the action of practitioners rather than being explicitly articulated, making it difficult to clearly express and document [25]. This inherent limitation reduces the effectiveness of traditional media, such as videos or written descriptions, to support comprehensive reflection [8].

As digital media continues to evolve, various sensing technologies have opened new possibilities for learning and reflection in craft practices. Motion capture (MoCap) can quantify spatial features of bodily postures, helping practitioners gain a better understanding of essential skill elements [4, 11, 19]. Electromyography (EMG) can record muscle activity and facilitate human motor skill learning [1, 18, 40]. Meanwhile, the sonification of craft gestures allows practitioners to grasp the nuances of their movements [35]. Collectively, these methods have enhanced both the depth and engagement of reflective processes of professional craft practices, offering craftsmen and learners richer sensory dimensions and data support. However, most of these methods rely primarily on visual or auditory channels. As the most direct sensory medium for craftsmen during the making process, there is little exploration of leveraging the tactile channel to support reflection for craft practices. As Sennett emphasizes, “*all skills, even the most abstract, begin with physical practice*” [34], bodily sensations such as touch, is critical in the development and refinement of craft skills. This embodied interactive process has not yet been fully reflected in existing reflection support systems.

To mitigate this gap, we collaborated with professional potters from Okinawa Prefecture in Japan, to develop a system that integrates FMG (Force-myography), accelerometer-based fingertip vibration sensing, and haptic cues. FMG is robust against sweat-induced noise in hot, humid workshops, making it well-suited to the local environment. By converting muscle activity and fingertip vibrations into haptic stimuli for post-action review, our system aims to bridge the gap between objective data and embodied reflection. Through a preliminary study, we found that combining haptic cues with visual information helps practitioners both deepen their skill understanding and communicate tacit techniques with added nuance.

2 Related Works

2.1 Convey the Tacit Knowledge in Craft Practice

In recent years, tacit knowledge has drawn increasing attention in craft and hybrid craft research [9, 10, 39] due to its embodied and often unspoken nature [25]. Traditional documentation approaches, such as photographs, videos, and journals, can visually capture operational details and experiences [29, 32, 33], facilitating reflection-in-action or reflection-on-action. Yet visual and textual records alone remain insufficient to convey the physical knowledge central to skill mastery [6, 8].

Following this line, researchers have started to look into how to utilize wearable sensors to record movements in real-time [18, 35]. For instance, EMG can help potters visualize muscle activity for post-action self-reflection [18], while accelerometers and magnetometers track subtle finger gestures in weaving, which has been explored in sonification for collective learning [35]. Meanwhile,

a “more-than-human centered design” perspective places sensing devices on both tools and materials (e.g., planes in woodworking), illuminating multi-dimensional human-tool-material interactions [24]. These data-capture strategies open new possibilities for enhancing our understanding of tacit knowledge.

2.2 Craft from an Embodied Perspective

Touch is fundamental in craft as it offers immediate, precise feedback that guides force and technique [34]. In pottery, subtle tactile cues inform makers about clay’s texture, elasticity, and moisture, enabling continuous refinements [2]. Material Engagement Theory (MET) characterizes craft as a co-creative process in which materials “speak back,” prompting adjustments as they sag or resist [20]. Here, tacit knowledge encompasses not just unspoken practice but the practitioner’s evolving perception of material properties. While some studies use accelerometers and vibrotactile replay to let users “re-experience” tactile sensations, such methods remain underexplored for reflective skill practice, indicating the viability of haptic-based feedback mechanisms for reflective practice.

2.3 The Potential of Haptic Technologies for Reflective Practice

For the purpose of social interaction, haptic technologies have been explored for sharing physical movements [17, 36]. Haptic systems that capture and reproduce movement can be used to understand other people’s muscle activities [15, 23]. These previous works demonstrate the potential of using haptic technologies to foster an understanding of the physical activities between two individuals, and therefore has the potential to apply to understanding muscle activities in craftsmanship.

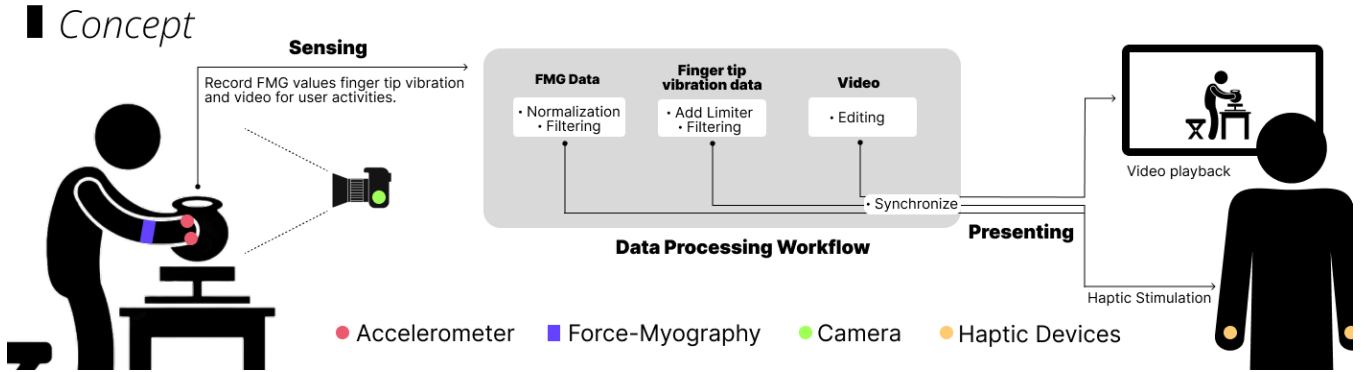
In traditional apprenticeship, both novices and masters rely on embodied demonstrations to develop their understanding of the specific action [7, 21]. By capturing and reintroducing visual, tactile, and muscle-activity data, modern haptic interfaces can augment hands-on knowledge transmission with interactive feedback that deepens reflection. Prior research shows that skin-stretch or vibrotactile interventions can enhance motor skills learning [12, 22]. We set to explore whether it can have a similar effect in the pottery setting.

3 EmbodyCraft

3.1 Concept

Drawing on the background discussions, we proposed *EmbodyCraft*, which uses recordings of muscle activity and fingertip vibrations to spark reflection. This includes three key data streams, muscle activity, fingertip vibrations, and video recordings—into a unified, time-synchronized dataset. We aim to use *EmbodyCraft* to capture and reproduce the nuanced interplay between the potter’s body, the material, and the craft process, thereby offering a richer medium for reflective practices in craft practices.

We focused on the wheel-throwing process [5] of pottery in this experience. Craft practices are often described as a “dialogue” between the practitioner and the material [3, 31]. In pottery making, the “mindful” or “attentive” movements of the practitioner

Figure 2: The concept framework of *EmbodyCraft*

respond to the material’s flows and resistances, creating a reciprocal exchange [14]. Compared to other crafts such as metalwork, woodworking, or glass replication, pottery demands a heightened sensitivity to the material’s responses, making it a powerful example of this dynamic interaction. Among the various stages of pottery, wheel throwing [30] – where the practitioner shapes the clay on a spinning wheel – stands out as a moment of profound engagement. This stage has been described as “an almost poetic moment of tacit knowledge at work”, where the potter intuitively adjusts their movements based on subtle tactile cues, exemplifying how body and material coalesce in the act of creation[27].

3.2 Hardware Implementation

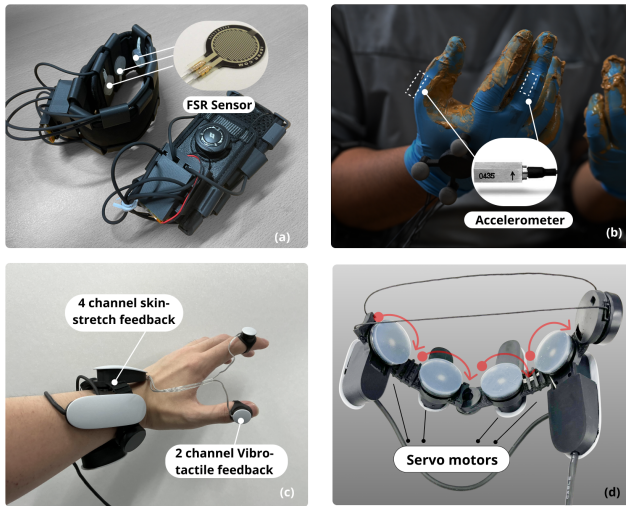


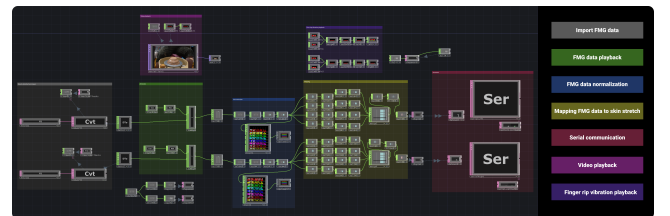
Figure 3: (a) Two FMG device with 8 channel FSR sensors (b) Accelerometer (c) FELLTECH wear - A haptic device combined 4 channels of skin-stretch stimulation and two channels of vibro-tactile (D) Distribution and direction of rotation of motors

To measure forearm muscle activity, we employed force myography (FMG) instead of traditional electromyography (EMG) due

to its higher robustness against sweat-induced noise in hot, humid workshop environments. Forearm muscle activity plays a pivotal role in coordinating hand and wrist movements, especially in tasks requiring both gripping and wrist forces[28]. In pottery, core actions such as throwing, pulling, and compressing clay demand similarly intricate coordination of the wrist and fingers, orchestrated by these forearm muscles. Ververidis et al. [38] further revealed that gestures which appear identical (e.g., “pulling up” vs. “circular downward shaping”) exhibit distinct muscle activation patterns when creating different vessel types—subtleties often overlooked through visual observation alone. By capturing these forearm nuances, our system aimed to uncover otherwise hidden force variations, and thus provide an exploratory avenue for deeper post-action reflection. Specifically, we integrated eight force-sensitive resistors (FSRs) into a custom adjustable wristband, evenly spaced around the forearm to maintain consistent spatial coverage. This arrangement ensured stable sensor contact and provided a structured distribution of data points for subsequent processing.

For fingertip tactile recording, we attached lightweight accelerometers, which capture subtle vibrations arising during clay manipulation. Haptic cues are delivered via the FeelTech Wear device [37], which features four channels of evenly spaced rotational skin-stretch motors —and two channels of vibrotactile actuators per hand. The skin-stretch stimuli reflect variations in muscle activity, while the vibrotactile actuators replay fingertip vibrations recorded by the accelerometers, collectively supporting a multimodal reflection on both muscular effort and tactile interactions.

3.3 Software Implementation

Figure 4: The control software of *EmbodyCraft*

As illustrated in Figure 4, we developed a multi-stream processing pipeline in TouchDesigner¹ to manage video playback, fingertip vibration output, and skin-stretch stimulation concurrently. One key objective was to enable real-time signal processing, allowing immediate adjustments of stimulation intensity based on user input or emerging insights. To map the sensor data to the four skin-stretch motors, we grouped every two adjacent sensors, averaged their outputs, and applied a linear mapping to the servomotor actuation range for skin-stretch stimuli. By allowing users to modify haptic parameters dynamically, our approach ensures both responsive and personalized reflective experiences, integrating visual and tactile cues seamlessly during post-action reflection.

4 Workshops

After finalizing our prototype, we organized two workshops with 6 potters in Okinawa, Japan in December 2024. We were inspired by previous works on reflective practice [31] and experiential learning cycle [16] in designing the workshop. These workshops aimed to explore how haptic stimulation derived from muscle activity (FMG) and fingertip vibrations might support reflective practice and skill communication in pottery-making.

Below, we detail each workshop's procedure and highlight **key findings**—illustrated with direct participant quotes—to reveal how tactile information shaped reflective practices in both self-review and observational learning.

4.1 Workshop 1: Post-action self-reflection on one's own data

4.1.1 Participant and procedure. To investigate how our system might enhance a potter's personal reflection-on-action, we began by working with a single expert. In this workshop, an experienced potter (S), specializing in wheel-throwing, served as our skill recorder. During the process, we captured forearm muscle activity (FMG), fingertip vibrations (accelerometers), and second-person video.

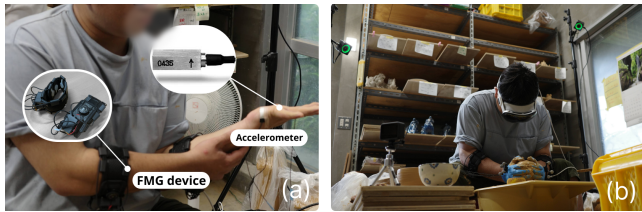


Figure 5: (a) Skill Recorder S with sensors attached. (b) Recording muscle activity, fingertip vibrations, and video during the wheel-throwing process.

Afterward, S revisited the same footage in two stages: (1) **Visual-Only** (baseline video), and (2) **Visual + Haptic** (video augmented by skin-stretch & vibrotactile playback derived from the recorded data). S performed a “Speak Along” protocol (talking aloud during viewing) and took part in a short interview at the end, discussing how the tactile cues influenced his reflection on skill details and force distribution.

¹<https://derivative.ca/>

4.1.2 Findings. Under the **visual-only** condition, S largely relied on the clay's visible changes, estimating force in a rough manner. He mentioned that “the amount of clay adhering to the hand can hint at force levels, but it's fairly coarse.” He also believed his left and right hands contributed force relatively evenly, primarily using the thumb, index, and middle fingers to control the clay. Thus, his reflection remained at a macro level: broad positioning of fingers and overall timing, without addressing fine-grained force distribution.

Once haptic cues were introduced, however, S noticed stronger-than-expected rotational signals on his right hand, while the left hand's fingertip vibrations appeared relatively weak. Contradicting his assumption of “balanced” force, he discovered—by correlating haptic cues with the video—that his right fingertips often overlapped or pressed atop the left, causing the right hand to exert more pressure than intended. As S put it, “I was applying force where I had no memory of it... and not applying force where I thought I was.”



Figure 6: (a) Skill recorder S experiencing the recorded data (b) Interview with Skill recorder S (c) Example of a discrepancy between SS's perceived “equal force” (blue) and the actual recorded force concentration (red) during wheel-throwing.

Figure 6(c) illustrates this discrepancy between S's perceived (balanced) and actually recorded force. Faced with this mismatch, S reconsidered his habitual hand positions, concluding that he was unconsciously relying more on the right hand. He likened the tactile cues to a “magnifying glass,” enabling him to pinpoint exactly when and where force became uneven. As a result, S devised more precise strategies, such as keeping the left hand more engaged and avoiding excessive fingertip overlap on the right side. In his words, “When there're tactile cues, it's like searching for mistakes—concentrating on where my force might be off.”

4.2 Workshop 2: Reflective observation and communication on other's data

4.2.1 Participants and Procedure. Building on our exploration of how haptic cues might enhance personal reflection (Workshop 1), we also sought to investigate whether such cues could facilitate deeper understanding and communication of skill data recorded by others. To this end, we organized a second workshop with five additional potters (“skill observers”), each having varying experience in wheel-throwing. They reviewed the same dataset captured by the skill recorder (S) in Workshop 1, again in two viewing stages: **Visual-Only (baseline video)**, and **Visual + Haptic** (video augmented by skin-stretch & vibrotactile cues).

As before, a “Speak Along” protocol prompted observers to articulate their thoughts during playback, and short interviews were conducted thereafter. Notably, the skill recorder S joined a final group discussion to clarify or comment on the observers’ interpretations, allowing us to examine how haptic cues shaped not only individual reflection but also collective communication about subtle craft techniques.

4.2.2 Findings. Without any haptic stimulation, these observers primarily focused on changes in the clay’s shape or the potter’s overall hand positioning. One observer noted how “finger marks and how quickly the clay deforms” might hint at force application, yet admitted it was “hard to pinpoint exactly how much force is used just by looking.” Consequently, feedback remained largely at a macro level, such as overall posture or when the potter was raising or compressing clay.

Once haptic cues were introduced, however, participants became more attuned to “exact force timings” and “patterns of increasing or decreasing pressure.” In one case, an observer expected strong force to stabilize a lopsided clay section but felt only weak rotational signals from the FMG-based skin-stretch stimulation: “I assumed he was pressing down really firmly, but the tactile signal was weak. Maybe he’s bracing with his elbow instead.”

Figure 7(c) illustrates this discrepancy. By correlating these tactile signals with the video, they discovered that S sometimes relied more on elbow support or subtle changes in hand angles, rather than overt forearm force. Observers remarked that haptic cues “fills in the blanks” about where and when precise force is applied, revealing tacit strategies not obvious from vision alone.

During the group discussion, S confirmed or corrected certain interpretations, explaining how he purposely minimized arm force at key moments. One observer commented, “With haptics, a whole new dimension appears—we see why he isn’t using as much force as we assumed.” This shared tactile reference made it easier to identify exactly which finger or motion phase merited attention. The conversation thus shifted from “broad guesswork” to a more precise exchange of subtle hand–material interactions, suggesting that haptic replay fosters richer communication of otherwise invisible force dynamics.

5 Discussion

5.1 Summary of Findings

Overall, our findings across the two workshops indicate that incorporating haptic replay into conventional video material offers a



Figure 7: (a) A Skill recorder “Y” experiencing the recorded data (b) Discussion with Skill observer “M” (c) Illustration of the discrepancy between perceived (blue) and actual (red) force application during clay manipulation

more refined glimpse into previously overlooked details of wheel-throwing. With only visual playback, participants focused on broad features—such as hand position or clay shape—leading to assumptions about constant or balanced force. However, the addition of haptic cues allowed them to pinpoint subtler variations—right-hand dominance, slight but crucial moments of lighter force, or reliance on elbow support—revealing that some supposed “even pressure” was actually more uneven. This heightened precision prompted deeper reflection among experienced potters, who noticed they were applying force where they had “no memory of it,” and it shifted observers’ discussions from vague estimates to targeted analysis of finger usage and timing. In short, haptic cues provided a shared reference point that fostered more nuanced dialogue and corrected misconceptions, suggesting that visual-plus-haptic review can significantly enhance the clarity and depth of skill reflection and communication.

5.2 Comparison with Related Work

Our preliminary findings partially resonate with the work of Smith et al. [35] and Liu et al. [18]. The former uses sonification of motion data to help craft practitioners discern subtle movement differences that might otherwise go unnoticed; in parallel, our tactile replay directly maps to muscle activity—particularly force distribution and force rhythm—as well as fingertip vibration, thereby strengthening practitioners’ embodied awareness. The latter employs EMG visualization to guide more in-depth reflection on action, whereas we opt for tactile cues instead of a purely visual display, employing a more embodied approach that enables users to physically sense the recorded data. The integration of video with multi-channel haptic cues enables craft practitioners to analyze skill execution

with greater granularity, facilitating the reassessment of prior assumptions and promoting in-depth reflection and discussion.

5.3 Limitations

While *EmbodyCraft* demonstrated potential for revealing subtle force dynamics in wheel-throwing, several limitations should be noted. First, our workshops involved a small cohort of experienced potters, limiting generalizability to novices or other crafts. Second, wearing multiple sensors can disrupt normal workflow, suggesting a need for further miniaturization and robust waterproofing for broader adoption. Third, participants always reviewed the visual-only playback before visual + haptic, so potential order effects were not fully isolated or controlled. Finally, participants required guidance to interpret the haptic cues (e.g., mapping skin-stretch intensities to muscle forces), implying a learning curve that may affect immediate usability in standard workshop settings.

6 Conclusion

In this paper, we introduced *EmbodyCraft*, a system that records forearm muscle activity and fingertip vibrations to generate haptic cues for post-action reflection in wheel-throwing. Through two workshops with professional potters, we found that combining visual playback with skin-stretch and vibrotactile cues revealed hidden force disparities, spurred deeper self-reflection, and facilitated more precise discussions of skill-based nuances. These initial findings highlight how replaying subtle embodied interactions with materials can enhance reflective practice and knowledge exchange in craft practices.

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